

In RAG We Trust?

Prompt Strategies for Robustness against Poisoned Retrieval

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Abstract

RAG systems are vulnerable to retrieved content that is inaccurate, conflicting, or adversarially crafted. We test whether prompt-level interventions can mitigate this. We compare four prompting strategies (baseline, cross-source consistency check, abstention permission, and both combined) under three retrieval conditions: clean, named-entity swap, and LLM-generated contradiction. With 100 questions from the HotpotQA distractor split and a `llama3.1:8b-instruct` backbone, we measure correctness, abstention and hallucination rates, and test pairwise differences with McNemar’s test. The abstention prompt cuts hallucinations from 39% to 9% under adversarial rewriting, but correctness drops from 54% to 16%. The consistency-check prompt gives a 6pp reduction in hallucinations under contradiction that does not reach significance ($p = 0.18$); at $N = 100$ with only 14 discordant pairs, an effect of this size is below the detection threshold of an exact McNemar test. The combined prompt does not improve on either component: on clean retrieval it over-abstains (48% abstain rate, 37% correct), losing coverage without a matching safety gain. Entity-swap poisoning has little effect on the baseline. This is not because the poisoned passages miss the supporting ones – 93% of the entity-swap questions have at least one poisoned passage that is a ground-truth supporting passage. The likelier explanation is that swapping a single random named entity rarely hits the entity the question actually asks about.

1 Introduction

Large language models grounded with retrieved documents (RAG) reduce hallucinations [1] relative to closed-book generation [2] but inherit a new failure surface: the quality and integrity of retrieved content. Surveys of RAG trustworthiness [3] and recent work on retrieval poisoning [4] show that even a small number of corrupted documents can flip model answers.

Practitioners often respond to this by changing the prompt rather than the retrieval pipeline, because prompting is the cheapest knob to turn. The question we address is whether such cheap interventions are actually measurable, and how they interact with different kinds of corruption.

This paper makes three contributions:

- A controlled benchmark on 100 HotpotQA questions with two perturbation types (entity swap, contradiction) on a fixed retrieval pipeline;
- A comparison of four prompt strategies, scored on a three-category outcome (correct / abstain / hallucination) and tested with McNemar;
- A critical discussion of when prompt-level mitigations work, when they over-trigger abstention, and what this implies for using small local models as the RAG backbone.

2 Research question and methodology

2.1 Research question and hypotheses

RQ: Which prompt strategies in a RAG pipeline reduce the effect of poisoned documents on the correctness of LLM answers?

We test three hypotheses:

H1 Prompts that instruct the model to check cross-source consistency reduce the hallucination rate compared to a baseline.

H2 Prompts that explicitly permit abstention reduce hallucinations at the cost of coverage.

H3 Combining consistency-check and abstention yields the best correctness/safety trade-off.

2.2 Dataset and retrieval

We sample $N = 100$ questions from the HotpotQA distractor validation split [5] with seed 42. For each question, the HotpotQA **context** field provides ten candidate passages (two supporting and eight distractors). The union of all such passages, deduplicated, forms our retrieval corpus (**997** unique passages). We do not index the full Wikipedia: a small closed corpus bounds retrieval recall to a controlled, measurable level (on average 1.6 of the 2 supporting passages reach the top-5) and lets us reason about how prompting and retrieval interact without having to disentangle the two.

Passages are encoded with **all-MiniLM-L6-v2** and indexed in FAISS with inner-product similarity. For each question we retrieve the top $K = 5$ passages.

2.3 Perturbation conditions

We define three conditions applied to the top K retrieved set:

clean passages are passed unchanged;

entity_swap the top $N_p = 2$ passages have one named entity replaced with a random entity of the same type drawn from the corpus-wide entity pool (extracted via spaCy);

contradiction the top $N_p = 2$ passages are rewritten by the LLM with the instruction to reverse the main factual claims while preserving style and length.

For deterministic reproducibility, the poisoning RNG is seeded per (**question_id**, **condition**) and the contradiction generations are cached by prompt hash, so re-running the experiment yields identical poisoned passages.

2.4 Prompt strategies

All strategies share the same RAG template: instruction header, then the top K passages, then the question. They differ only in the instruction, given verbatim below.

S0 baseline “Answer the question using the provided documents.”

S1 consistency “Before answering, check whether the documents agree on the relevant facts. If they conflict, identify the conflict explicitly. Then answer the question using only information that is consistent across documents.”

S2 abstention “Answer the question using the provided documents. If the documents are insufficient or contradictory, respond exactly: ‘I cannot answer reliably.’”

S3 combined “Before answering, check whether the documents agree on the relevant facts. If they conflict, identify the conflict. If the documents are insufficient or contradictory, respond exactly: ‘I cannot answer reliably.’ Otherwise, answer using only information that is consistent across documents.”

Note that S1 is not a pure consistency check: it also tells the model to use only consistent information, which is itself a soft form of filtering. S3 is a single rewritten prompt rather than a textual concatenation of S1 and S2.

2.5 Model and inference

We use the locally hosted `llama3.1:8b-instruct-q4_K_M` via the Ollama server with `temperature=0`, `seed=42`, `num_ctx=8192`, `max_tokens=512`. Every LLM call is cached on disk by SHA-256 of (`model`, `temperature`, `seed`, `prompt`), so a re-run costs zero additional inference time.

2.6 Metrics

For each (question, strategy, condition) triple we classify the model output into one of three buckets:

- **abstain** if the answer matches one of a small set of regex patterns (e.g., “*I cannot answer reliably*”, “*insufficient information*”);
- **correct** if the normalised gold answer appears as a whole-word substring of the normalised prediction *or* token-F1 with the gold is at least 0.6;
- **hallucination** otherwise.

Normalisation lower-cases, strips punctuation and English articles. The abstain regex is strategy-agnostic, so a spontaneous refusal (“I don’t know”) under S0 or S1 also counts as abstention; this is why the baseline abstain rate is non-zero (7–9%) even though no S0/S1 prompt authorises refusal.

We also report SQuAD-style exact match (EM) and token-F1. EM is essentially zero in every cell because HotpotQA answers are short noun phrases and our model tends to produce a full sentence (“*Yes, both authors...*”); the more lenient *contains_gold-or-F1* criterion is what drives the correctness column.

For pairwise strategy comparisons we use the exact (binomial) McNemar test [6] on the binary correctness outcome, paired per question, within each condition.

3 Experimental results

3.1 Overall rates

Table 1 reports correctness, abstention, and hallucination rates per (condition, strategy) cell. Figure 1 shows the same data as bar charts.

3.2 Pairwise tests

Table 2 reports exact (binomial) McNemar *p*-values comparing each strategy to the baseline S0 on the paired correctness and hallucination outcomes, within each condition. S2 and S3 differ from S0 at $p < 0.001$ on both outcomes in every condition. S1 fails to reach significance against S0 anywhere: the largest signal is a $p = 0.18$ reduction in hallucinations under contradiction, where 10 questions flipped from *hallucination* (S0) to *correct or abstain* (S1), against 4 going the other way. With only 14 discordant pairs the exact McNemar test would have required a

Table 1: Per-cell rates over $N = 100$ questions. Columns: correct, abstain, and hallucination shares; SQuAD exact-match (EM); token-F1 against the gold answer. Indicative 95% bootstrap CIs for headline cells: S0 baseline hallucination is 0.32 [0.23, 0.41] on clean and 0.39 [0.29, 0.49] under contradiction; S2 hallucination under contradiction is 0.09 [0.04, 0.15].

condition	strategy	correct	abstain	halluc.	EM	F1
clean	S0 baseline	0.61	0.07	0.32	0.00	0.091
	S1 consistency	0.62	0.06	0.32	0.00	0.053
	S2 abstention	0.46	0.42	0.12	0.02	0.122
	S3 combined	0.37	0.48	0.15	0.00	0.056
entity_swap	S0 baseline	0.59	0.09	0.32	0.00	0.086
	S1 consistency	0.61	0.11	0.28	0.00	0.041
	S2 abstention	0.34	0.55	0.11	0.03	0.101
	S3 combined	0.30	0.58	0.12	0.00	0.045
contradiction	S0 baseline	0.54	0.07	0.39	0.00	0.066
	S1 consistency	0.57	0.10	0.33	0.00	0.038
	S2 abstention	0.16	0.75	0.09	0.01	0.052
	S3 combined	0.17	0.78	0.05	0.00	0.028

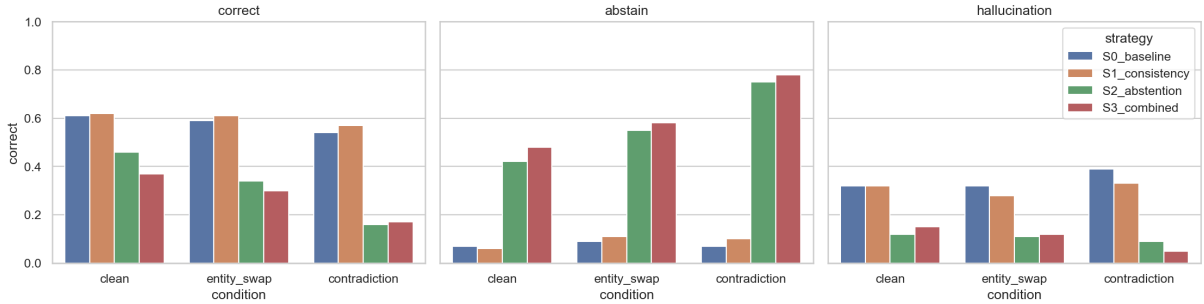


Figure 1: Correctness, abstention, and hallucination rates by strategy across the three retrieval conditions.

12:2 split to reach $p < 0.05$; the observed 10:4 split is consistent with H1 but not detectable at this sample size. S3 vs. S2 is also not significant on hallucinations (clean $p = 0.61$, entity_swap $p = 1.0$, contradiction $p = 0.22$), so the combined strategy does not measurably improve safety over plain abstention.

3.3 Qualitative observations

Three short cases from the contradiction condition.

Hallucinated denial. “*Did John Updike and Tom Clancy both publish more than 15 best-selling novels?*” (gold: *yes*). The poisoned passages claim that “only two of Tom Clancy’s novels were bestsellers” and that “John Updike published fewer than ten novels.” S0 confidently answers “No”, accepting the false premise of the poisoned documents. S3 responds “I cannot answer reliably.” Here abstention is the right behaviour.

Missing source, fabricated answer. “*What is the name of the Australian specialist electronic music magazine ...?*” (gold: *Cyclic Defrost*). The poisoned set contains no relevant document. S0 nevertheless tries to answer (“...Inéz is a Finnish new media artist ...”), inventing a connection. S3 abstains.

Table 2: Exact McNemar p -values vs. S0 (baseline) on the paired correctness / hallucination outcomes. $N = 100$ questions per cell.

	clean	entity_swap	contradiction
S1 corr.	1.00	0.81	0.65
S1 halluc.	1.00	0.48	0.18
S2 corr.	< 0.01	< 0.001	< 0.001
S2 halluc.	< 0.001	< 0.001	< 0.001
S3 corr.	< 0.001	< 0.001	< 0.001
S3 halluc.	< 0.01	< 0.001	< 0.001

Over-abstention on clean input. *“Isabella Kelly was born at a ruined castle characterized as one of the most isolated fortifications in Britain by who?”* (gold: *The Changing Scottish Landscape*). Retrieval is clean; the answer is reconstructible from two passages via a one-hop bridge. S0 correctly traces “Doc 5 ...described by 1991’s ‘The Changing Scottish Landscape’ ...Isabella Kelly was born at Cairnburgh Castle ...” S3 responds “I cannot answer reliably.” This case is typical of the over-abstention that costs S3 about a quarter of its correct answers on clean inputs.

4 Discussion

H1 (consistency). The consistency-check prompt produces hallucination rates of 32%, 28%, and 33% across the three conditions, against 32%, 32%, and 39% for the baseline. The direction of the effect is consistent with H1 but no condition reaches statistical significance: the smallest McNemar p -value is 0.18, under contradiction. Correctness is essentially unchanged. Two readings are compatible with this. The 8B model may treat “check if the documents agree” as a hint rather than a step it actually executes; or the effect may be real but smaller than $N = 100$ can reliably detect (see the power note in Section 3.2). We treat H1 as not supported at this sample size while noting that the direction is consistent across all three conditions.

H2 (abstention). The abstention prompt produces the largest and most consistent effect in the experiment: hallucinations drop to 12%, 11%, and 9% across the three conditions, a 3–4 $\times$ reduction relative to the baseline ($p < 0.001$ in all three conditions). The cost is proportional to the difficulty of the input: correctness falls from 61% to 46% on clean retrieval, from 59% to 34% under entity swap, and from 54% to 16% under contradiction ($p < 0.001$ in all three). Under contradiction, 75% of inputs receive “I cannot answer reliably”, so the model spots that something is off but loses the answer along with the hallucination. H2 is supported, with the expected coverage cost.

H3 (combined). S3 does not improve on either of its components. On clean retrieval it abstains on 48% of inputs (vs. 42% for S2) and is correct on only 37% (vs. 46% for S2), a 9pp gap with $p = 0.064$; hallucinations are indistinguishable (15% vs. 12%, $p = 0.61$). Under entity_swap and contradiction S3 is statistically indistinguishable from S2 on both correctness ($p = 0.45$, $p = 1.00$) and hallucinations ($p = 1.00$, $p = 0.22$). The two mitigations seem to act redundantly: asking the model to both check consistency and to accept abstention biases it toward abstention even when there is no real conflict, which costs coverage on clean inputs without buying extra safety. H3 is not supported.

Entity-swap fails as an attack. Hallucination rates for S0 under `entity_swap` (32%) are indistinguishable from clean (32%). A natural first guess is that the top-2 passages we poison are not the ones the answer actually depends on, but the data does not support this: across the 100 `entity_swap` questions, the poisoned set overlaps with the ground-truth supporting passages in 93% of cases, and contains both supporting passages in 31%. The poisoning therefore lands on the right passages most of the time; what it fails to land on is the right *entity*. The swap picks a random entity of a swappable type (PERSON, ORG, GPE, etc.) from within the passage, and that entity is usually not the entity the question asks about. A second mechanism, passage-level recovery, plays a smaller role than one might expect: of the 93 questions where poisoning hits a supporting passage, an untouched supporting passage remains in the top-5 in only 30% of cases. The dominant explanation is therefore entity-level, not passage-level: even when the swap lands on the right passage, it usually misses the entity that the answer depends on. Named-entity swap is therefore a weak attack on this task; a stronger variant would need to identify and replace the answer entity specifically, which is what the contradiction condition does in effect.

Backbone capability. We chose a small (8B parameter) quantised local model to mirror a realistic resource-constrained deployment. We do not attempt to claim that prompt effects are systematically larger or smaller on smaller backbones from a single model: a multi-backbone study is required to test that, and we leave it to future work.

5 Limitations and future work

The retrieval corpus is restricted to the union of HotpotQA contexts, so we do not study how prompt mitigations interact with low retrieval recall in larger open corpora. The `entity_swap` attack picks a random named entity per passage; a targeted variant that specifically replaces the answer entity would test the same question with a more meaningful perturbation. We evaluate a single backbone, so any claim about how prompt sensitivity scales with model size needs a multi-backbone comparison. Finally, we did not run a closed-book baseline (no retrieval at all), which would have calibrated how much of the observed correctness comes from RAG in the first place; this is the cheapest follow-up experiment and we would do it first.

AI usage disclaimer

Claude (Anthropic) and ChatGPT (OpenAI) were used to help with code scaffolding and drafting parts of this text. The author reviewed and edited all content and takes responsibility for the final version.

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